[CSL252]-Design and Analysis of Algorithms **2025-26-W**

TUTORIAL VIII

Date of the tutorial: **Mar 12, 2026**.

Due Date: **11:59 pm, Mar 16, 2026**.

Instructions. Same as Tutorial I.

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1. TRUE/FALSE: Let G be a graph with n vertices and m edges.
   (a) All its DFS forests (for traversals starting at different vertices) will have the same number of tree edges and the same number of back edges?
   (b) All its DFS forests will have the same number of trees?
2. TRUE/FALSE:
   (a) If a directed graph G contains a path from u to v and if $u.d < v.d$ in a depth-first search of G, then v is a descendant of u in the depth-first forest produced.
   (b) A vertex u of a directed graph can end up in a depth-first tree containing only u, even though u has both incoming and outgoing edges in G.
   (c) If a directed graph G contains a path from u to v, then any depth-first search must result in $v.d \leq u.f$.
3. Give a linear-time algorithm to check whether a given directed acyclic graph G contain a directed path that touches (contains) every vertex exactly once.
4. You are given a directed acyclic graph $G = (V, E)$ in which each node $u \in V$ has an associated price, denoted by $price(u)$, which is a positive integer. The cost of a node u, denoted by $cost(u)$, is defined to be the price of the cheapest node reachable from u (including u itself). Design an algorithm that computes $cost(u)$ for all $u \in V$. Write the pseudocode.
5. Recall that a directed graph G is strongly connected if, for any two vertices u and v, there is a path in G from u to v and a path in G from v to u. Describe an algorithm to determine, given an undirected connected graph G as input, whether it is possible to direct each edge of G so that the resulting directed graph is strongly connected.
6. The police department in the city of Delhi has made every street in the city one-way. Despite widespread complaints from confused motorists, the mayor claims that it is possible to legally drive from any intersection in Delhi to any other intersection.
   (a) The city needs to either verify or refute the mayor's claim. Formalize this problem in terms of graphs, and then describe and analyze an algorithm to solve it.
   (b) After running your algorithm from part (a), the mayor reluctantly admits that she was misinformed.
   Suppose now that the mayor needs to show that, even though her original claim was false, something weaker does hold: If you start navigating one-way streets from airport, there is always a way to drive legally back to the airport. Formulate this weaker property as a graph-theoretic problem and show that this can also be solved in linear time.

7. There are n rebels that are currently planning, well, a "party". A rebel v can send messages to only one other rebel, designated as $contact(v)$. Here, if a rebel is given a message, they will send it to their contact, and it would be propagated in this fashion to everyone reachable.

   (a) Describe an algorithm, as fast as possible, that computes a rebel v, such that if you send a message to v, the message gets propagated to all the rebels. If there is not such vertex v, the algorithm should output "no solution".

   (b) Describe an algorithm, as fast as possible, that computes the minimal number of rebels that needs to be sent directly a message, before the message can be propagated to all the rebels.